

A Novel Ferroelectric FET-Based Adaptively-Stochastic Neuron for Stimulated-Annealing Based Optimizer With Ultra-Low Hardware Cost

Jin Luo^{ID}, Graduate Student Member, IEEE, Tianyi Liu, Zhiyuan Fu, Xinming Wei, Mengxuan Yang, Liang Chen, Student Member, IEEE, Qianqian Huang^{ID}, Member, IEEE, and Ru Huang, Fellow, IEEE

Abstract—In this work, based on ferroelectric FET (FeFET), a novel capacitor-less and bio-inspired adaptively-stochastic neuron is proposed and experimentally demonstrated for the first time for solving optimization problems in spiking neural network with remarkably reduced hardware cost. By exploiting the physics of inherent stochastic dynamic process of ferroelectric domains nucleation, the proposed FeFET-based neuron with only three transistors can realize adaptively stochastic spike firing behavior, where its stochasticity gradually decreases during operation. The adaptive stochasticity is experimentally demonstrated for the hardware implementation of stochastic simulated annealing algorithm for optimization, providing a promising ultralow-hardware-cost solution for solving optimization problems.

Index Terms—Ferroelectric FET (FeFET), stochastic spiking neuron, stimulated annealing, optimization problem.

I. INTRODUCTION

COMBINATORIAL optimization problems (CSPs) that belong to NP-hard class are ubiquitous in energy-starved edge AI applications [1], [2] like automotive logistics and scheduling systems. Inspired by human brain, the spiking neural networks (SNNs) operating with stochastic simulated annealing (SSA) algorithm (Fig. 1a) allow to find the solutions of complex CSPs with high energy efficiency [3]. For hardware implementation of SSA in SNN, adaptively-stochastic neuron

Manuscript received December 16, 2021; accepted December 23, 2021. Date of publication December 27, 2021; date of current version January 27, 2022. This work was supported in part by the National Key Research and Development Program of China under Grant 2018YFB2202801; in part by the NSFC under Grant 61927901, Grant 61822401, Grant 61851401, and Grant 61421005; and in part by the 111 Project under Grant B18001. The review of this letter was arranged by Editor S. Yu. (Corresponding authors: Qianqian Huang; Ru Huang.)

Jin Luo, Tianyi Liu, Zhiyuan Fu, Xinming Wei, Mengxuan Yang, and Liang Chen are with the Key Laboratory of Microelectronic Devices and Circuits (MOE), School of Integrated Circuits, Peking University, Beijing 100871, China.

Qianqian Huang and Ru Huang are with the Key Laboratory of Microelectronic Devices and Circuits (MOE), School of Integrated Circuits, Peking University, Beijing 100871, China, also with the Beijing Laboratory of Future IC Technology and Science, Peking University, Beijing 100871, China, and also with the Chinese Institute for Brain Research (CIBR), Beijing 102206, China (e-mail: hqq@pku.edu.cn; ruhuang@pku.edu.cn).

Color versions of one or more figures in this letter are available at <https://doi.org/10.1109/LED.2021.3138765>.

Digital Object Identifier 10.1109/LED.2021.3138765

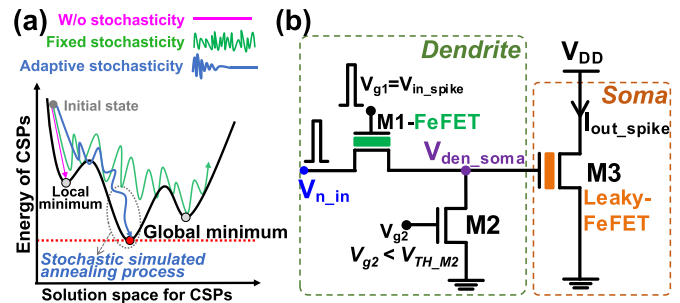


Fig. 1. (a) The adaptive-stochasticity-based stochastic simulated annealing (SSA) process when solving combinatorial optimization problems (CSPs) in spiking neural network (SNN). (b) The proposed FeFET-based adaptively-stochastic neuron.

where its stochasticity gradually decreases during operation is necessary for escaping from local minima and gradually converging to global minimum as shown in Fig. 1a. By far only CMOS spiking neurons with complex sophisticated periphery circuits like random number generators and control circuits for randomness decay have been implemented for the adaptive stochasticity of SSA in SNN [2]–[4], yet suffering from high-hardware-cost issue.

In this work, we propose and experimentally demonstrate a novel capacitor-less adaptively-stochastic neuron with only three transistors based on CMOS-compatible ferroelectric FET (FeFET) (Fig. 1b). The neuronal dendrite is designed for dynamic amplitude modulation of postsynaptic signal, and then the neuronal soma generates stochastic spikes adaptively. The proposed FeFET-based neuron is experimentally demonstrated with adaptively decreased stochasticity for the hardware implementation of SSA algorithm with ultra-low hardware cost. This letter is organized as follows. Section II explains the design of FeFET based adaptively-stochastic neuron. Section III presents the experimental results of FeFETs characteristics and adaptively stochastic behavior of the proposed neuron. Finally, the conclusion is presented in Section IV.

II. ADAPTIVELY-STOCHASTIC NEURON DESIGN

Fig. 1b shows the proposed adaptively-stochastic neuron circuit topology based on FeFET devices without the need of capacitors. The neuronal dendrite consists a normal FeFET (M1) and a conventional NMOS (M2). The dendrite receives input pulse (V_{in_spike}) from pre-neuron, and the V_{n_in} has the

relatively lower amplitude which is generated by modulating V_{in_spike} with synapse module. V_{in_spike} and V_{n_in} are synchronous considering the negligible synaptic delay. The neuronal soma is designed based on our proposed novel leaky-FeFET (L-FeFET) [7]–[9], which can emulate basic integration, leak and reset properties by polarization accumulation with accelerated degradation.

When synchronously receiving V_{in_spike} and the postsynaptic signal V_{n_in} with lower amplitude at gate and drain of normal FeFET (M1) respectively, the feedforward transmission transistor M1 is turned on with positive pulse of V_{in_spike} and will transmit the postsynaptic pulses to the gate of soma L-FeFET with threshold-voltage-loss effect in transmission process. M2 is designed to be biased in the subthreshold region to discharge the soma nodal capacitance without affecting the transmission process of M1. Meanwhile, M1 is initially reset to relatively high threshold voltage (V_{TH}) state before the SNN starts operation, without frequent program operation in SSA-based SNN and relaxing the requirement of endurance for device M1. Owing to the gate voltage pulse, the FE polarization of M1 will be gradually switched and retained, resulting in the gradually decreased V_{TH} of M1. Therefore, the amplitude of pulse signals transmitted to soma L-FeFET (V_{den_soma}) will increase during overall operational process of SSA-based SNN.

Due to the accumulative polarization reversal with voltage pulse applied and the leaky effect, L-FeFET is promising to realize basic leaky integrate-and-fire (LIF) and self-reset function, which have already been demonstrated in our previous work [7], and the partial polarization switching is utilized, alleviating the endurance issue of FeFET. On the other hand, the fundamental dynamic stochastic process of domain nucleation may lead to the switching voltage and time fluctuation for ferroelectric domains [10], and thus for the L-FeFET based neuron, the number of pulses for firing is stochastic when repeating the neuronal LIF cycles. Besides, the variation of ferroelectric polarization switching dynamics decreases when the applied voltage of pulses increases [11]. Accordingly, the stochasticity of spike generated by soma L-FeFET will adaptively decrease due to the increased applied voltage of intermediate pulse signal (V_{den_soma}). The drain terminal of M3 is a universal design of neuronal output port, which may also need additional component when connecting with diverse designs of artificial synapse. Therefore, the proposed FeFET-based neuron can adaptively implement the decreasing stochasticity during operation for the solution of CSPs in SSA-based SNNs.

III. EXPERIMENTAL DEMONSTRATION

A. FeFETs Characteristics for Adaptively-Stochastic Neuron

In this experiment, both $Hf_{0.5}Zr_{0.5}O_2$ (HZO) films are adopted for the L-FeFET and normal FeFET fabrications. Fig. 2a shows the experimental I_d - V_g curves of normal FeFET. Different from the conventional FeFET, for the realization of L-FeFET, the HZO layer is designed with partial crystallization phase to properly increase its leakage current for leaky effect as we described in [7]. Fig. 2b and c show the measured dynamics of fabricated normal FeFET. As for normal FeFET, the device channel conductance gradually rises (/falls) with positive (/negative) voltage pulses applied to the gate (Fig. 2b), indicating V_{TH} modulation due to polarization switching and

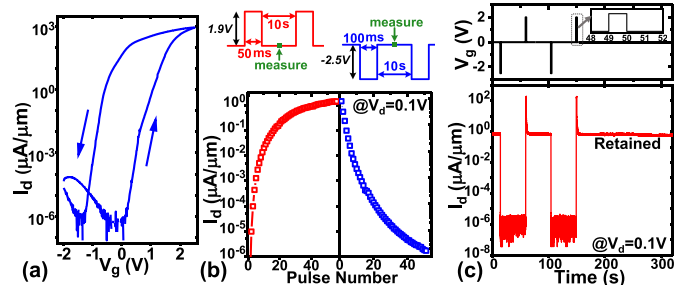


Fig. 2. Experimental characterization of normal FeFET (M1) constructed by connecting metal-ferroelectric-metal (MFM) structure to the gate of foundry baseline MOSFET. The $Hf_{0.5}Zr_{0.5}O_2$ layer is grown by atomic layer deposition with annealing temperature of 450° , whose area is $40 \times 40 \mu m^2$. The gate length and width of the baseline MOSFET are $0.13 \mu m$ and $0.28 \mu m$, respectively. (a) Measured I_d - V_g curves. (b) The channel conductance modulation with positive/negative V_g pulses applied, and (c) long-term retention characteristics.

its non-volatile behavior. The modulated channel conductance of FeFET can be retained for a long time after program/erase (Fig. 2c), which is different from the leaky property of accelerated polarization degradation in L-FeFET [7].

B. Adaptively-Stochastic Neuron Behavior

Fig. 3a shows the measured basic LIF function with stochasticity of proposed neuron. When the normal FeFET (M1) of dendrite is sufficiently conductive, L-FeFET (M3) can accumulate excitatory input pulses due to accumulated FE polarization, causing that the V_{TH} of M3 gradually decreases. When the accumulation of switching probability under pulse trains is high [11], [12] and the V_{TH} is low enough, the voltage of next input pulse makes the M3 channel sufficiently conductive, resulting the output spike firing. Since the required pulse number for firing is not determinate but probabilistic with the same input pulse train, stochastic behavior is demonstrated in our fabricated neuron, and the statistics of number of accumulated pulses required for firing shows Gaussian distribution as shown in Fig. 3a. The width of voltage pulse in the measurement depends on the polarization switching speed of the fabricated ferroelectric layer, and can be further reduced by optimizing the area or thickness [15].

As shown in Fig. 3b, from the initial high V_{TH} state, when the feedforward transmission transistor FeFET (M1) receives synchronous input pulse trains applied to both drain and gate from pre-neurons, it can pass the input pulse signal (V_{n_in}) to the intermediate node (V_{den_soma}) with threshold-voltage-loss effect, and M2 is biased in the subthreshold region to release the stored charge in circuit nodal capacitance of M3 gate node and recover V_{den_soma} to 0V. Simultaneously, the V_{TH} of normal FeFET (M1) will gradually decrease due to the accumulated FE polarization reversal caused by voltage pulses at the device gate. Therefore, the voltage amplitude of intermediate signals applied to L-FeFET soma gradually increase with the stimulation time.

Fig. 3c shows the statistically experimental results of L-FeFET soma behavior. The standard deviations (STD, σ) of number of pulses for firing gradually decrease with increased V_{den_soma} (Fig. 3d), which means that the stochasticity of proposed neuron decreases with gate voltage of L-FeFET soma increasing. The neuronal dendrite and neuronal soma are separately measured in order to show dynamics of the

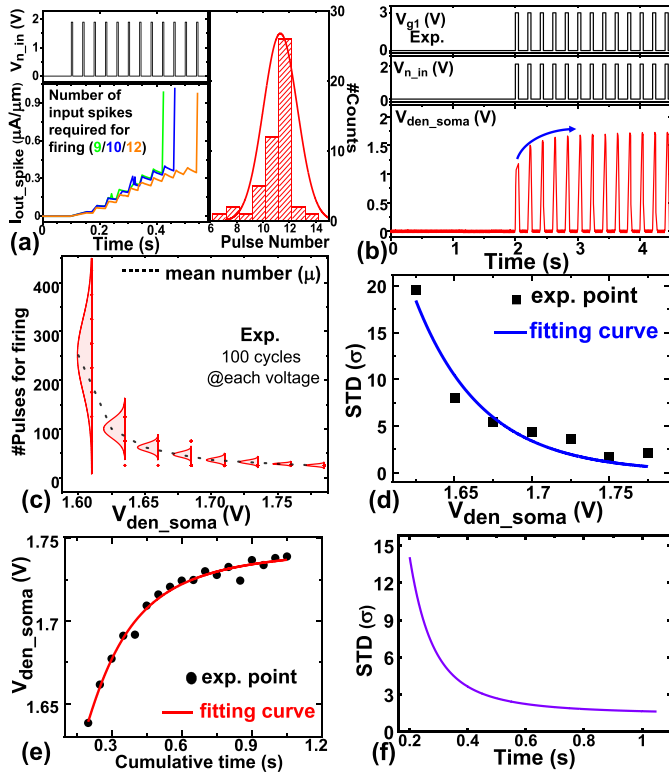


Fig. 3. Measured behavior of neuronal soma and dendrite based on the breadboard-based circuit made from single devices. (a) Measured LIF behavior and stochastic behavior of proposed neuron based on fabricated single M3 device as we described in [7]. The number of input pulses required to fire the neuronal soma is probabilistic, which shows Gaussian distribution. (b) Measured dendrite behavior based on cascade of M1 and M2 transistors on breadboard. The dynamic modulation of V_{den_soma} with neuronal dendrite when the input spikes are applied to V_{n_in} and V_{g1} . The M2 is biased in subthreshold region, equivalent to a resistor. (c) Measured statistical deviations (STD, σ) of number of pulses for firing with increased V_{den_soma} based on M3 device, and (d) relation between STD (σ) and V_{den_soma} . (e) Relation between V_{den_soma} and cumulative time. (f) The adaptively-decreasing stochasticity of proposed FeFET-based neuron can be captured according to (d) and (e).

intermediate signal in the adaptively-stochastic neuron circuit. The relationship between V_{den_soma} and cumulative time, and the relationship between STD (σ) and V_{den_soma} are extracted and fitted based on experimental results as shown in Fig. 3e and d, respectively. Then the dynamic process of STD (σ) over cumulative stimulation time can be obtained (Fig. 3f). It can be seen that the neuronal stochasticity of spike firing behavior (STD, σ) can adaptively decrease with cumulative input stimulus time increasing. Moreover, the stochasticity of the proposed neuron can also be regulated by adjusting the pulse width due to the time-dependent stochasticity of polarization switching [11]. Based on the proposed FeFET neuron, the SNN enabling SSA is further studied for Sudoku CSP solving [4] (Fig. 4), which takes adaptively stochastic neurons for each number grid of Sudoku. The annealing process of SSA algorithm is achieved by utilizing the experimental evolution process of neuronal stochasticity as shown in Fig. 4a. The solving process in Fig. 4b shows that due to the adaptively stochastic neuron design, the systems can escape from the local minima and can finally converge to the optimal solution, realizing the SSA algorithm for solving optimization problems.

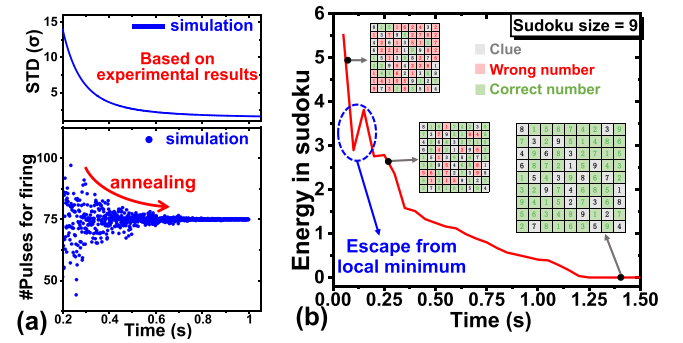


Fig. 4. The proposed FeFET neuron based SNN for Sudoku CSP solving with SSA. (a) The evolution process of neuronal stochasticity in SSA is set as experimental results from Fig. 3f, and the simulated result shows annealing process. (b) The solving process with escaping from local minimum.

TABLE I

BENCHMARK OF THE PROPOSED ADAPTIVELY-STOCHASTIC NEURON FOR CSP SOLVING WITH OTHER REPORTED RESULTS FOR CSPs

	Ref. [14]	Ref. [13]	Ref. [6]	This work
Working mode	Clock-driven	Event-driven		Event-driven
Physics for neuron	CMOS	Capacitor charging		Polarization Switching
Neuronal hardware cost	2 Sense-amplifiers + 1 Comparator	3C+29T	3OPAMPs + 1C3R + Logic	3T
Physics for stochasticity	Noise of memristor read-out current	Analog circuit mismatch	Stochastic crystallization of PCM	Stochastic dynamics of FE nucleation
Adaptive stochasticity for annealing	No (Require peripheral control circuit)			Yes

Table I compares the proposed adaptively-stochastic neuron with other reported neurons for CSP solving. By utilizing the physics of ferroelectric (FE) gradually accumulative polarization switching to emulate the neuronal membrane potential, large area capacitor can be eliminated in our proposed design. Besides, by utilizing nonvolatile V_{TH} modulation of normal FeFET and intrinsic stochastic dynamics of FE nucleation, adaptively stochastic neuron is implemented by only three transistors without peripheral adaptive control circuit. It indicates that the proposed neuron can realize SSA algorithm in SNN for optimization task.

IV. CONCLUSION

This work proposed and experimentally demonstrated a novel FeFET-based adaptively-stochastic neuron with ultralow hardware cost of only three transistors. Based on the design, the neuronal gradual decreased stochasticity during neuronal operation is implemented for stochastic stimulated-annealing algorithm, which can realize escaping from local minima and gradually converging to global minimum in SNNs based optimizer, showing its great potential for ultralow-hardware-cost neuromorphic system.

ACKNOWLEDGMENT

The authors thank Yue Peng and Prof. Genquan Han for HZO deposition, and thank High-performance Computing Platform of Peking University for computing support.

REFERENCES

- [1] M. Yamaoka, C. Yoshimura, M. Hayashi, T. Okuyama, H. Aoki, and H. Mizuno, "A 20 k-spin Ising chip to solve combinatorial optimization problems with CMOS annealing," *IEEE J. Solid-State Circuits*, vol. 51, no. 1, pp. 303–309, Jan. 2016, doi: 10.1109/JSSC.2015.2498601.

- [2] Y. Su, H. Kim, and B. Kim, "CIM-Spin: A 0.5-to-1.2 V scalable annealing processor using digital compute-in-memory spin operators and register-based spins for combinatorial optimization problems," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2020, pp. 480–482, doi: [10.1109/ISSCC19947.2020.9062938](https://doi.org/10.1109/ISSCC19947.2020.9062938).
- [3] Z. Jonke, S. Habenschuss, and W. Maass, "Solving constraint satisfaction problems with networks of spiking neurons," *Frontiers Neurosci.*, vol. 10, p. 118, Mar. 2016, doi: [10.3389/fnins.2016.00118](https://doi.org/10.3389/fnins.2016.00118).
- [4] G. A. F. Guerra and S. B. Furber, "Using stochastic spiking neural networks on SpiNNaker to solve constraint satisfaction problems," *Frontiers Neurosci.*, vol. 11, p. 714, Dec. 2017, doi: [10.3389/fnins.2017.00714](https://doi.org/10.3389/fnins.2017.00714).
- [5] M. Davies, N. Srinivasa, T. H. Lin, G. Chinya, Y. Cao, S. H. Choday, G. Dimou, P. Joshi, N. Imam, S. Jain, Y. Liao, C. K. Lin, A. Lines, R. Liu, D. Mathaikutty, S. McCoy, A. Paul, J. Tse, G. Venkataramanan, Y. H. Weng, A. Wild, Y. Yang, and H. Wang, "Loihi: A neuromorphic manycore processor with on-chip learning," *IEEE Micro*, vol. 38, no. 1, pp. 82–99, Jan. 2018, doi: [10.1109/MM.2018.112130359](https://doi.org/10.1109/MM.2018.112130359).
- [6] G. Pedretti, P. Mannocci, S. Hashemkhani, V. Milo, O. Melnic, E. Chicca, and D. Ielmini, "A spiking recurrent neural network with phase-change memory neurons and synapses for the accelerated solution of constraint satisfaction problems," *IEEE J. Explor. Solid-State Comput. Devices Circuits*, vol. 6, no. 1, pp. 89–97, Jun. 2020, doi: [10.1109/JXDC.2020.2992691](https://doi.org/10.1109/JXDC.2020.2992691).
- [7] C. Chen, M. Yang, S. Liu, T. Liu, K. Zhu, Y. Zhao, H. Wang, Q. Huang, and R. Huang, "Bio-inspired neurons based on novel leaky-FeFET with ultra-low hardware cost and advanced functionality for all-ferroelectric neural network," in *Proc. 39th Symp. VLSI Technol.*, Jun. 2019, pp. T136–T137, doi: [10.23919/VLSIT.2019.8776495](https://doi.org/10.23919/VLSIT.2019.8776495).
- [8] J. Luo, L. Yu, T. Liu, M. Yang, Z. Fu, Z. Liang, L. Chen, C. Chen, S. Liu, S. Wu, Q. Huang, and R. Huang, "Capacitor-less stochastic leaky-FeFET neuron of both excitatory and inhibitory connections for SNN with reduced hardware cost," in *IEDM Tech. Dig.*, Dec. 2019, pp. 6.4.1–6.4.4, doi: [10.1109/IEDM19573.2019.8993535](https://doi.org/10.1109/IEDM19573.2019.8993535).
- [9] T. Liu, J. Luo, X. Wei, Q. Huang, and R. Huang, "A novel leaky-FeFET based true random number generator with ultralow hardware cost for neuromorphic application," in *Proc. 5th IEEE Electron Devices Technol. Manuf. Conf. (EDTM)*, Apr. 2021, pp. 1–3, doi: [10.1109/EDTM50988.2021.9420857](https://doi.org/10.1109/EDTM50988.2021.9420857).
- [10] H. Mulaosmanovic, T. Mikolajick, and S. Slesazeck, "Random number generation based on ferroelectric switching," *IEEE Electron Device Lett.*, vol. 39, no. 1, pp. 135–138, Jan. 2018, doi: [10.1109/LED.2017.2771818](https://doi.org/10.1109/LED.2017.2771818).
- [11] S. Deng, G. Yin, W. Chakraborty, S. Dutta, S. Datta, X. Li, and K. Ni, "A comprehensive model for ferroelectric FET capturing the key behaviors: Scalability, variation, stochasticity, and accumulation," in *Proc. 40th Symp. VLSI Technol.*, Jun. 2020, pp. TF1.4.1–TF1.4.2, doi: [10.1109/VLSITechnology18217.2020.9265014](https://doi.org/10.1109/VLSITechnology18217.2020.9265014).
- [12] C. Cheng, P. Tiw, Y. Cai, X. Yan, Y. Yang, and R. Huang, "In-memory computing with emerging nonvolatile memory devices," *Sci. China Inf. Sci.*, vol. 64, no. 12, pp. 1–46, Nov. 2021, doi: [10.1007/s11432-021-3327-7](https://doi.org/10.1007/s11432-021-3327-7).
- [13] H. Mostafa, L. K. Müller, and G. Indiveri, "An event-based architecture for solving constraint satisfaction problems," *Nature Commun.*, vol. 6, no. 1, Dec. 2015, Art. no. 8941, doi: [10.1038/ncomms9941](https://doi.org/10.1038/ncomms9941).
- [14] M. R. Mahmoodi, H. Kim, Z. Fahimi, H. Nili, L. Sedov, V. Polishchuk, and D. B. Strukov, "An analog neuro-optimizer with adaptable annealing based on 64×64 OT1R crossbar circuit," in *IEDM Tech. Dig.*, Dec. 2019, pp. 14.7.1–14.7.4, doi: [10.1109/IEDM19573.2019.8993442](https://doi.org/10.1109/IEDM19573.2019.8993442).
- [15] X. Lyu, M. Si, P. R. Shrestha, K. P. Cheung, and P. D. Ye, "First direct measurement of sub-nanosecond polarization switching in ferroelectric hafnium zirconium oxide," in *IEDM Tech. Dig.*, Dec. 2019, pp. 15.2.1–15.2.4, doi: [10.1109/IEDM19573.2019.8993509](https://doi.org/10.1109/IEDM19573.2019.8993509).